

# COMPUTER SCIENCE

## Overview

<http://www.cs.iastate.edu>

The department of Computer Science offers Bachelor of Science and Bachelor of Arts degrees in Computer Science in the College of Liberal Arts and Sciences.

The undergraduate curriculum in Computer Science leading to the Bachelor of Science degree is accredited by the Computing Accreditation Commission of ABET, <http://www.abet.org>. This degree equips students with a sound knowledge of the foundations of Computer Science as well as problem-solving and system design skills necessary to create robust, efficient, reliable, scalable, and flexible software systems. The B.S. degree in Computer Science prepares students for graduate study in Computer Science and for various business, industry, and government positions including computer scientists, information technologists, and software developers.

The purpose of the Bachelor of Arts degree in Computer Science is to accommodate students who wish to pursue a major in Computer Science while taking more courses outside of the major than a Bachelor of Science degree program permits. This will include students pursuing double majors, students pursuing one or more strong minors, and students pursuing broader liberal educations. The B.A. program retains the rigor of our B.S. program while allowing students to spread the breadth of their education over other disciplines in addition to Computer Science.

The main educational objectives of the B.S. degree program in Computer Science at Iowa State University are that its graduates demonstrate expertise, engagement, and learning within two to five years after graduation.

- **Expertise:** Graduated students should have the ability to establish peer-recognized expertise in the discipline. They should have the ability to articulate this expertise by formulating and solving problems of interest, by creating or deriving value through the application of technology, and by using mathematical foundations, algorithmic principles, and computer science theory in designing, implementing and evaluating computer-based systems and processes which meet the desired needs of their employers.
- **Engagement:** Graduated students should have the ability to be engaged in the profession through the practice of computer science in industry, academia, or the public sector. They should demonstrate effective teaming and commitment to working with others by applying communications skills and professional knowledge.
- **Learning:** Graduated students should have the ability to engage in sustained learning through graduate work, professional improvement

opportunities, and self-study so that they can adapt to the role played by information processing in ever-changing areas of science, technology, and society.

## Student Learning Outcomes

Upon graduation, students should have:

1. An ability to analyze a complex computing problem, and to apply principles of computing and other relevant disciplines to identify solutions.
2. An ability to design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. An ability to communicate effectively in a variety of professional contexts.
4. An ability to recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. An ability to function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. An ability to apply computer science theory and software development fundamentals to produce computing-based solutions.

## Undergraduate Curriculum in Software Engineering

The Department of Computer Science, together with the Department of Electrical and Computer Engineering, also offer a curriculum leading to an undergraduate degree in Software Engineering (<https://catalog.iastate.edu/collegeofliberalartsandsciences/softwareengineering/>). The Software Engineering curriculum offers emphasis areas in Software Engineering principles, process, and practice. Students may also take elective courses in Computer Engineering and Computer Science.

## Degree Requirements

As majors in the College of Liberal Arts and Sciences, students pursuing both the B.A. and the B.S. in Computer Science must meet College of Liberal Arts and Sciences (<https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext>) and University-wide requirements (<https://catalog.iastate.edu/collegescurricula/>) for graduation in addition to those stated below for the major.

LAS majors require a minimum of 120 credits, including a minimum of 45 credits at the 3000/4000 level. You must also complete the LAS world language requirement and career proficiency requirement.

Students in all ISU majors must complete a three-credit course in U.S. Cultures and Communities and a three-credit course in International Perspectives. Check (<http://www.registrar.iastate.edu/courses/div-ip->

guide.html) for a list of approved courses. Discuss with your advisor how the two courses that you select can be applied to your graduation plan.

### Degree requirements for the B.S. in Computer Science:

The B.S. departmental requirements consist of a minimum of 49 credits in Computer Science and satisfaction of written and oral requirements.

The following courses are required:

|            |                                                                     |   |
|------------|---------------------------------------------------------------------|---|
| COMS 1010  | Orientation                                                         | R |
| COMS 1270  | Introduction to Computer Programming                                | 3 |
| COMS 2270  | Object-oriented Programming                                         | 4 |
| COMS 2280  | Introduction to Data Structures                                     | 3 |
| COMS 3090  | Software Development Practices                                      | 3 |
| COMS 3110  | Introduction to the Design and Analysis of Algorithms               | 3 |
| COMS 3210  | Introduction to Computer Architecture and Machine-Level Programming | 3 |
| COMS 3270  | Advanced Programming Techniques                                     | 3 |
| COMS 3310  | Theory of Computing                                                 | 3 |
| COMS 3420  | Principles of Programming Languages                                 | 3 |
| COMS 3520  | Introduction to Operating Systems                                   | 3 |
| COMS 4020C | Computer Science Senior Project: Project in Computer Science        | 3 |

At least 15 credits, including at least 6 credits of 4000-level courses, 15 all with a grade of C- or better, from the following:

|           |                                                                        |
|-----------|------------------------------------------------------------------------|
| COMS 3190 | Construction of User Interfaces                                        |
| COMS 3360 | Introduction to Computer Graphics                                      |
| COMS 3620 | Object-Oriented Analysis and Design                                    |
| COMS 3630 | Introduction to Database Management Systems                            |
| COMS 4070 | Applied Formal Methods                                                 |
| COMS 4090 | Software Requirements Engineering                                      |
| COMS 4120 | Formal Methods in Software Engineering                                 |
| COMS 4130 | Program Analysis                                                       |
| COMS 4150 | Software System Safety                                                 |
| COMS 4170 | Software Testing                                                       |
| COMS 4180 | Introduction to Computational Geometry                                 |
| COMS 4190 | Trustworthy Healthcare Software                                        |
| COMS 4210 | Logic for Mathematics and Computer Science                             |
| COMS 4240 | Introduction to High Performance Computing                             |
| COMS 4250 | High Performance Computing for Scientific and Engineering Applications |
| COMS 4260 | Introduction to Parallel Algorithms and Programming                    |
| COMS 4330 | Molecular Programming of Nanoscale Devices and Processes               |

|            |                                                            |
|------------|------------------------------------------------------------|
| COMS 4340  | Quantum Information and Complexity                         |
| COMS 4350  | Algorithms for Large Data Sets: Theory and Practice        |
| COMS 4370  | Computer Game and Media Programming                        |
| COMS 4400  | Principles and Practice of Compiling                       |
| COMS 4410  | Programming Languages                                      |
| COMS 4530  | Privacy Preserving Algorithms and Data Security            |
| COMS 4540  | Distributed Systems                                        |
| COMS 4550  | Simulation: Algorithms and Implementation                  |
| COMS 4590  | Security and Privacy in Cloud Computing                    |
| COMS 4720  | Principles of Artificial Intelligence                      |
| COMS 4740  | Introduction to Machine Learning                           |
| COMS 4760  | Motion Planning for Robotics and Autonomous Systems        |
| COMS 4762X | Introduction to Mobile Robotics                            |
| COMS 4770  | Foundations of Robotics and Computer Vision                |
| COMS 4810  | Numerical Methods for Differential Equations               |
| COMS 4870  | Introduction to Network Programming and Cloud Computing    |
| COMS 4880  | Computer Networks                                          |
| CPRE 4300  | Network Protocols and Security                             |
| CPRE 4310  | Basics of Information System Security                      |
| CPRE 4580  | Real Time Systems                                          |
| CPRE 4890  | Computer Networking and Data Communications                |
| SE 4160    | Software Evolution and Maintenance                         |
| SE 4190    | Software Tools for Large Scale Data Analysis               |
| SE 4210    | Software Analysis and Verification for Safety and Security |
| SE 4220    | Cloud Computing - Software Development                     |

**Total Credits** **49**

Toward satisfying the requirements of the College of Liberal Arts and Sciences, the following courses should be included:

|                                            |                                                 |    |
|--------------------------------------------|-------------------------------------------------|----|
| PHIL 3430                                  | Ethics of Computing and Artificial Intelligence | 3  |
| SPCM 2120                                  | Fundamentals of Public Speaking                 | 3  |
| LAS 2030                                   | Professional Career Preparation                 | 1  |
| At least 17 credits of Math and Statistics |                                                 | 17 |
| MATH 1650                                  | Calculus I                                      | 4  |
| MATH 1660                                  | Calculus II                                     | 4  |
| COMS 2300                                  | Discrete Computational Structures               | 3  |
| One Statistics course from:                |                                                 |    |
| STAT 3005                                  | Engineering Statistics                          | 3  |
| STAT 3030                                  | Probability and Statistics for Computer Science | 3  |

|           |                                                            |   |
|-----------|------------------------------------------------------------|---|
| STAT 3441 | Introduction to the Theory of Probability and Statistics I | 4 |
|-----------|------------------------------------------------------------|---|

|           |                                                     |   |
|-----------|-----------------------------------------------------|---|
| STAT 3447 | Probability and Statistical Theory for Data Science | 4 |
|-----------|-----------------------------------------------------|---|

At least one Math course from:

|           |                             |   |
|-----------|-----------------------------|---|
| MATH 2070 | Matrices and Linear Algebra | 3 |
|-----------|-----------------------------|---|

|           |              |   |
|-----------|--------------|---|
| MATH 2650 | Calculus III | 4 |
|-----------|--------------|---|

|           |                                   |   |
|-----------|-----------------------------------|---|
| MATH 2660 | Elementary Differential Equations | 3 |
|-----------|-----------------------------------|---|

|           |                                                          |   |
|-----------|----------------------------------------------------------|---|
| MATH 2670 | Elementary Differential Equations and Laplace Transforms | 4 |
|-----------|----------------------------------------------------------|---|

|           |               |   |
|-----------|---------------|---|
| MATH 3040 | Combinatorics | 3 |
|-----------|---------------|---|

|           |              |   |
|-----------|--------------|---|
| MATH 3140 | Graph Theory | 3 |
|-----------|--------------|---|

|           |                          |   |
|-----------|--------------------------|---|
| MATH 3170 | Theory of Linear Algebra | 4 |
|-----------|--------------------------|---|

One of the following 2-course Natural Science sequences (with labs):

|           |                         |   |
|-----------|-------------------------|---|
| BIOL 2110 | Principles of Biology I | 8 |
|-----------|-------------------------|---|

|         |                                        |  |
|---------|----------------------------------------|--|
| & 2110L | and Principles of Biology Laboratory I |  |
|---------|----------------------------------------|--|

|             |                              |  |
|-------------|------------------------------|--|
| & BIOL 2120 | and Principles of Biology II |  |
|-------------|------------------------------|--|

|              |                                         |  |
|--------------|-----------------------------------------|--|
| & BIOL 2120L | and Principles of Biology Laboratory II |  |
|--------------|-----------------------------------------|--|

Or

|           |                               |   |
|-----------|-------------------------------|---|
| BIOL 2550 | Fundamentals of Human Anatomy | 8 |
|-----------|-------------------------------|---|

|         |                                              |  |
|---------|----------------------------------------------|--|
| & 2550L | and Fundamentals of Human Anatomy Laboratory |  |
|---------|----------------------------------------------|--|

|             |                                      |  |
|-------------|--------------------------------------|--|
| & BIOL 2560 | and Fundamentals of Human Physiology |  |
|-------------|--------------------------------------|--|

|              |                                                 |  |
|--------------|-------------------------------------------------|--|
| & BIOL 2560L | and Fundamentals of Human Physiology Laboratory |  |
|--------------|-------------------------------------------------|--|

Or

|           |                     |   |
|-----------|---------------------|---|
| CHEM 1770 | General Chemistry I | 9 |
|-----------|---------------------|---|

|         |                                       |  |
|---------|---------------------------------------|--|
| & 1770L | and Laboratory in General Chemistry I |  |
|---------|---------------------------------------|--|

|             |                          |  |
|-------------|--------------------------|--|
| & CHEM 1780 | and General Chemistry II |  |
|-------------|--------------------------|--|

|              |                                        |  |
|--------------|----------------------------------------|--|
| & CHEM 1780L | and Laboratory in College Chemistry II |  |
|--------------|----------------------------------------|--|

Or

|           |                     |   |
|-----------|---------------------|---|
| GEOL 1000 | How the Earth Works | 8 |
|-----------|---------------------|---|

|         |                                     |  |
|---------|-------------------------------------|--|
| & 1000L | and How the Earth Works: Laboratory |  |
|---------|-------------------------------------|--|

|             |                          |  |
|-------------|--------------------------|--|
| & GEOL 2020 | and History of the Earth |  |
|-------------|--------------------------|--|

|              |                                      |  |
|--------------|--------------------------------------|--|
| & GEOL 2020L | and History of the Earth: Laboratory |  |
|--------------|--------------------------------------|--|

Or

|           |                                     |    |
|-----------|-------------------------------------|----|
| PHYS 2310 | Introduction to Classical Physics I | 10 |
|-----------|-------------------------------------|----|

|         |                                                    |  |
|---------|----------------------------------------------------|--|
| & 2310L | and Introduction to Classical Physics I Laboratory |  |
|---------|----------------------------------------------------|--|

|             |                                          |  |
|-------------|------------------------------------------|--|
| & PHYS 2320 | and Introduction to Classical Physics II |  |
|-------------|------------------------------------------|--|

|              |                                                     |  |
|--------------|-----------------------------------------------------|--|
| & PHYS 2320L | and Introduction to Classical Physics II Laboratory |  |
|--------------|-----------------------------------------------------|--|

The following courses meet the communication proficiency requirement:

|          |                                        |   |
|----------|----------------------------------------|---|
| LIB 1600 | Introduction to College Level Research | 1 |
|----------|----------------------------------------|---|

|           |                                     |   |
|-----------|-------------------------------------|---|
| ENGL 1500 | Critical Thinking and Communication | 3 |
|-----------|-------------------------------------|---|

|           |                                                   |   |
|-----------|---------------------------------------------------|---|
| ENGL 2500 | Written, Oral, Visual, and Electronic Composition | 3 |
|-----------|---------------------------------------------------|---|

One of the following

|           |                        |   |
|-----------|------------------------|---|
| ENGL 3020 | Business Communication | 3 |
|-----------|------------------------|---|

|           |                              |   |
|-----------|------------------------------|---|
| ENGL 3050 | Creative Writing: Nonfiction | 3 |
|-----------|------------------------------|---|

|           |                             |   |
|-----------|-----------------------------|---|
| ENGL 3090 | Proposal and Report Writing | 3 |
|-----------|-----------------------------|---|

|           |                         |   |
|-----------|-------------------------|---|
| ENGL 3140 | Technical Communication | 3 |
|-----------|-------------------------|---|

## Degree requirements for the B.A. in Computer Science:

The B.A. departmental requirements consist of a minimum of 31 credits in Computer Science and satisfaction of written and oral requirements.

The following courses are required:

Computer Science core:

|           |             |   |
|-----------|-------------|---|
| COMS 1010 | Orientation | R |
|-----------|-------------|---|

|           |                                      |   |
|-----------|--------------------------------------|---|
| COMS 1270 | Introduction to Computer Programming | 3 |
|-----------|--------------------------------------|---|

|           |                             |   |
|-----------|-----------------------------|---|
| COMS 2270 | Object-oriented Programming | 4 |
|-----------|-----------------------------|---|

|           |                                 |   |
|-----------|---------------------------------|---|
| COMS 2280 | Introduction to Data Structures | 3 |
|-----------|---------------------------------|---|

|           |                                |   |
|-----------|--------------------------------|---|
| COMS 3090 | Software Development Practices | 3 |
|-----------|--------------------------------|---|

|           |                                                       |   |
|-----------|-------------------------------------------------------|---|
| COMS 3110 | Introduction to the Design and Analysis of Algorithms | 3 |
|-----------|-------------------------------------------------------|---|

|                                                                                                                                                                                                                                                   |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Advanced Computer Science, five 3000/4000-level courses chosen from lists (a) and (b) below, with at least one at the 4000-level and at least two chosen from list (a). COMS 4140 may not be applied towards fulfilling the 4000-level electives. | 15 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|

(a) Software and Hardware Systems:

|           |                                                                     |
|-----------|---------------------------------------------------------------------|
| COMS 3210 | Introduction to Computer Architecture and Machine-Level Programming |
|-----------|---------------------------------------------------------------------|

|           |                                 |
|-----------|---------------------------------|
| COMS 3270 | Advanced Programming Techniques |
|-----------|---------------------------------|

|           |                                     |
|-----------|-------------------------------------|
| COMS 3420 | Principles of Programming Languages |
|-----------|-------------------------------------|

|           |                                   |
|-----------|-----------------------------------|
| COMS 3520 | Introduction to Operating Systems |
|-----------|-----------------------------------|

|           |                                     |
|-----------|-------------------------------------|
| COMS 3620 | Object-Oriented Analysis and Design |
|-----------|-------------------------------------|

|           |                                             |
|-----------|---------------------------------------------|
| COMS 3630 | Introduction to Database Management Systems |
|-----------|---------------------------------------------|

|           |                                   |
|-----------|-----------------------------------|
| COMS 4090 | Software Requirements Engineering |
|-----------|-----------------------------------|

|           |                                        |
|-----------|----------------------------------------|
| COMS 4120 | Formal Methods in Software Engineering |
|-----------|----------------------------------------|

|           |                  |
|-----------|------------------|
| COMS 4130 | Program Analysis |
|-----------|------------------|

|           |                        |
|-----------|------------------------|
| COMS 4150 | Software System Safety |
|-----------|------------------------|

|           |                  |
|-----------|------------------|
| COMS 4170 | Software Testing |
|-----------|------------------|

|           |                                      |
|-----------|--------------------------------------|
| COMS 4400 | Principles and Practice of Compiling |
|-----------|--------------------------------------|

|           |                       |
|-----------|-----------------------|
| COMS 4410 | Programming Languages |
|-----------|-----------------------|

|           |                     |
|-----------|---------------------|
| COMS 4540 | Distributed Systems |
|-----------|---------------------|

|           |                                                         |
|-----------|---------------------------------------------------------|
| COMS 4870 | Introduction to Network Programming and Cloud Computing |
|-----------|---------------------------------------------------------|

|           |                   |
|-----------|-------------------|
| COMS 4880 | Computer Networks |
|-----------|-------------------|

(b) Other Advanced Computer Science:

|           |                                 |
|-----------|---------------------------------|
| COMS 3190 | Construction of User Interfaces |
|-----------|---------------------------------|

|            |                                                                        |
|------------|------------------------------------------------------------------------|
| COMS 3310  | Theory of Computing                                                    |
| COMS 3360  | Introduction to Computer Graphics                                      |
| COMS 3500  | Number Theory                                                          |
| COMS 4070  | Applied Formal Methods                                                 |
| COMS 4180  | Introduction to Computational Geometry                                 |
| COMS 4190  | Trustworthy Healthcare Software                                        |
| COMS 4210  | Logic for Mathematics and Computer Science                             |
| COMS 4240  | Introduction to High Performance Computing                             |
| COMS 4250  | High Performance Computing for Scientific and Engineering Applications |
| COMS 4260  | Introduction to Parallel Algorithms and Programming                    |
| COMS 4330  | Molecular Programming of Nanoscale Devices and Processes               |
| COMS 4340  | Quantum Information and Complexity                                     |
| COMS 4350  | Algorithms for Large Data Sets: Theory and Practice                    |
| COMS 4370  | Computer Game and Media Programming                                    |
| COMS 4530  | Privacy Preserving Algorithms and Data Security                        |
| COMS 4550  | Simulation: Algorithms and Implementation                              |
| COMS 4590  | Security and Privacy in Cloud Computing                                |
| COMS 4720  | Principles of Artificial Intelligence                                  |
| COMS 4740  | Introduction to Machine Learning                                       |
| COMS 4760  | Motion Planning for Robotics and Autonomous Systems                    |
| COMS 4762X | Introduction to Mobile Robotics                                        |
| COMS 4770  | Foundations of Robotics and Computer Vision                            |
| COMS 4810  | Numerical Methods for Differential Equations                           |

**Total Credits** **31**

Toward satisfying the requirements of the College of Liberal Arts and Sciences, the following courses should be included:

Mathematics Core (3 courses):

|           |                                   |
|-----------|-----------------------------------|
| MATH 1650 | Calculus I                        |
| MATH 1660 | Calculus II                       |
| COMS 2300 | Discrete Computational Structures |

Additional Mathematics (1 course from the following):

|           |                                                          |
|-----------|----------------------------------------------------------|
| COMS 3310 | Theory of Computing                                      |
| MATH 2070 | Matrices and Linear Algebra                              |
| MATH 2650 | Calculus III                                             |
| MATH 2660 | Elementary Differential Equations                        |
| MATH 2670 | Elementary Differential Equations and Laplace Transforms |

|                                                                |                                                            |
|----------------------------------------------------------------|------------------------------------------------------------|
| MATH 3040                                                      | Combinatorics                                              |
| MATH 3140                                                      | Graph Theory                                               |
| MATH 3170                                                      | Theory of Linear Algebra                                   |
| Statistics (1 course from the following):                      |                                                            |
| STAT 3005                                                      | Engineering Statistics                                     |
| STAT 3030                                                      | Probability and Statistics for Computer Science            |
| STAT 3441                                                      | Introduction to the Theory of Probability and Statistics I |
| STAT 3447                                                      | Probability and Statistical Theory for Data Science        |
| SPCM 2120                                                      | Fundamentals of Public Speaking                            |
| Natural Sciences: minimum 8 credits from approved College list |                                                            |

The following courses meet the communication proficiency requirement:

|                      |                                                   |   |
|----------------------|---------------------------------------------------|---|
| LIB 1600             | Introduction to College Level Research            | 1 |
| ENGL 1500            | Critical Thinking and Communication               | 3 |
| ENGL 2500            | Written, Oral, Visual, and Electronic Composition | 3 |
| One of the following |                                                   |   |
| ENGL 3020            | Business Communication                            | 3 |
| ENGL 3050            | Creative Writing: Nonfiction                      | 3 |
| ENGL 3090            | Proposal and Report Writing                       | 3 |
| ENGL 3140            | Technical Communication                           | 3 |

### The following pertains to both the B.S. and the B.A. in Computer Science:

Students must earn at least a C- in MATH 1650, MATH 1660, and each Computer Science course taken to fulfill the Degree Program. The LAS College requires the major must contain at least 8 credits in courses taken at Iowa State University that are numbered 3000 or above and in which the student's grade is C or higher.

According to the university-wide Communication Proficiency Grade Requirement (<https://catalog.iastate.edu/academics/#communicationproficiencytext>), students must demonstrate their communication proficiency by earning a grade of C or better in ENGL 2500. The Department requires a C or higher in the upper-level ENGL course (3020, 3050, 3090, 3140).

To obtain a bachelor's degree from the College of Liberal Arts and Sciences, curriculum in liberal arts and sciences, a student must earn at least 45 credits at the 3000 level or above taken at a four-year college. All such credits, including courses taken on a pass/not pass basis, may be used to meet this requirement.

Students must take at least 15 credits of Computer Science courses at the 3000 level or higher at Iowa State University while resident here.

Computer Science transfer courses need to be a minimum grade of C or higher to be considered for course substitution.

## Four Year Plan Computer Science, B.S.

### Freshman

| Fall           | Credits Spring      | Credits   |
|----------------|---------------------|-----------|
| COMS 1010      | R COMS 2270         | 4         |
| COMS 1270      | 3 MATH 1660         | 4         |
| MATH 1650      | 4 ENGL 2500         | 3         |
| ENGL 1500      | 3 LIB 1600          | 1         |
| SOCIAL SCIENCE | 3 ARTS & HUMANITIES | 3         |
| <b>13</b>      |                     | <b>15</b> |

### Sophomore

| Fall                             | Credits Spring                       | Credits      |
|----------------------------------|--------------------------------------|--------------|
| COMS 2280                        | 3 COMS 3210                          | 3            |
| COMS 2300                        | 3 COMS 3110                          | 3            |
| SCIENCE SEQUENCE PART 1          | 4 COMS 3000/4000 ELECTIVE            | 3            |
| SOCIAL SCIENCE                   | 3 SCIENCE SEQUENCE PART 2            | 4            |
| WORLD LANGUAGE 1010/<br>ELECTIVE | 3-4 WORLD LANGUAGE 1020/<br>ELECTIVE | 3-4          |
| LAS 2030                         | 1                                    |              |
| <b>17-18</b>                     |                                      | <b>16-17</b> |

### Junior

| Fall                    | Credits Spring       | Credits   |
|-------------------------|----------------------|-----------|
| COMS 3090               | 3 COMS 3270          | 3         |
| COMS 3000/4000 ELECTIVE | 3 COMS 3310          | 3         |
| MATH ELECTIVE           | 3 STAT 3000 ELECTIVE | 3         |
| SPCM 2120               | 3 ENGL 3000 ELECTIVE | 3         |
| ARTS & HUMANITIES       | 3 PHIL 3430          | 3         |
| <b>15</b>               |                      | <b>15</b> |

### Senior

| Fall                     | Credits Spring                   | Credits   |
|--------------------------|----------------------------------|-----------|
| COMS 3420                | 3 COMS 3520                      | 3         |
| COMS 3000/4000 ELECTIVE  | 3 COMS 4020                      | 3         |
| COMS 4000 ELECTIVE       | 3 COMS 4000 ELECTIVE             | 3         |
| SOCIAL SCIENCE & USCC/IP | 3 ARTS & HUMANITIES &<br>USCC/IP | 3         |
| ELECTIVE                 | 2-3 ELECTIVE                     | 3         |
| <b>14-15</b>             |                                  | <b>15</b> |

**Total Credits: 120-123**

## Computer Science, B.A.

### Freshman

| Fall           | Credits Spring        | Credits   |
|----------------|-----------------------|-----------|
| COMS 1010      | R COMS 2270           | 4         |
| COMS 1270      | 3 MATH 1660           | 4         |
| MATH 1650      | 4 ENGL 2500           | 3         |
| ENGL 1500      | 3 Arts and Humanities | 3         |
| Social Science | 3 LIB 1600            | 1         |
| <b>13</b>      |                       | <b>15</b> |

### Sophomore

| Fall              | Credits Spring        | Credits      |
|-------------------|-----------------------|--------------|
| COMS 2280         | 3 COMS 3110           | 3            |
| COMS 2300         | 3 Arts and Humanities | 3            |
| Natural Science 1 | 4 Natural Science 2   | 4            |
| Elective          | 3 Elective            | 3            |
| World Languages 1 | 3-4 World Languages 2 | 3-4          |
| LAS 2030          | 1                     |              |
| <b>17-18</b>      |                       | <b>16-17</b> |

### Junior

| Fall                | Credits Spring                     | Credits   |
|---------------------|------------------------------------|-----------|
| COMS 3090           | 3 COMS 3000/4000-level<br>Elective | 3         |
| MATH Elective       | 3-4 ENGL 3000-level Elective       | 3         |
| SPCM 2120           | 3 Elective                         | 3         |
| Arts and Humanities | 3 STAT 3000-level Elective         | 3         |
| Elective            | 3 Social Science                   | 3         |
| <b>15-16</b>        |                                    | <b>15</b> |

### Senior

| Fall                             | Credits Spring                       | Credits   |
|----------------------------------|--------------------------------------|-----------|
| COMS 3000/4000-level<br>Elective | 3 COMS 3000/4000-level<br>Elective   | 3         |
| COMS 3000/4000-level<br>Elective | 3 COMS 3000/4000-level<br>Elective   | 3         |
| Social Science and USCC/IP       | 3 Arts and Humanities and<br>USCC/IP | 3         |
| Elective                         | 3 Elective                           | 3         |
| Elective                         | 2-3 Elective                         | 3         |
| <b>14-15</b>                     |                                      | <b>15</b> |

**Total Credits: 120-124**

Note: The B.A. in Computer Science, like other B.A. degrees in LAS, gives students more freedom than the B.S. to design their own curricula. However, LAS degree requirements impose some rigor on this freedom. For example, the B.A. requires 24 upper-level credits in Computer Science,

Mathematics, and Statistics, but LAS requires at least 45 upper-level credits, so a substantial fraction of B.A. students' extra freedom will be confined to upper-level courses.

## Undergraduate Minor in Computer Science

The Department of Computer Science offers an undergraduate minor in Computer Science. The minor requires at least 16 credits in computer science courses.

A minimum grade of C- is required in COMS 2270 and COMS 2280. A minimum grade of C is required in both COMS 3110 and the three credits of 3000-level Computer Science courses and above. Students must meet all prerequisites for Computer Science courses taken to fulfill the minor. At least 6 credits of the minor must be in courses numbered 3000 and above with a grade of C or higher and 3 credits must be completed at ISU. The minor must include at least 3 credits that are not used to meet any other department, college, or university requirement.

|                                                      |                                                       |   |
|------------------------------------------------------|-------------------------------------------------------|---|
| COMS 2270                                            | Object-oriented Programming                           | 4 |
| COMS 2280                                            | Introduction to Data Structures                       | 3 |
| COMS 2300                                            | Discrete Computational Structures                     | 3 |
| COMS 3110                                            | Introduction to the Design and Analysis of Algorithms | 3 |
| 3 credits in COMS courses at the 3000 level or above |                                                       | 3 |

The Computer Science Undergraduate Minor is an LAS Minor. In addition to University policies governing minors (<https://catalog.iastate.edu/academics/#degreeplanningtext>), LAS minors require at least 6 credits in courses numbered 3000 and above, with a grade of C or higher.

## Certificate in Computing Applications

Students interested in the computing applications, may want to explore the Certificate in Computing Applications (<https://catalog.iastate.edu/collegeofliberalartsandsciences/computingapplicationscertificate/>); it is a cross-disciplinary course of study in the Colleges of Liberal Arts and Sciences, Engineering, and Business.

## Concurrent Bachelors and Masters Degrees

The Computer Science Department offers the following concurrent B.S./M.S. degree programs:

B.S./M.S. in Computer Science

B.S. in Computer Science /M.S. in Artificial Intelligence

B.S. in Software Engineering /M.S. in Computer Science

B.S. in Software Engineering /M.S. in Artificial Intelligence

These concurrent B.S./M.S. degrees offer highly motivated and focused students the opportunity for accelerated study. Students will be allowed to double count up to 12 credits. This program will pair undergraduate students in the concurrent program with research teams of graduate students and dedicated faculty members prominent in their fields of expertise. Students who declare concurrent enrollment will be considered graduate-level students, and therefore eligible for graduate research assistantships and scholarships. Click here for application instructions (<https://www.cs.iastate.edu/concurrent-bsms-computer-science/>).

More information can be found here: <https://www.cs.iastate.edu/cs.iastate.edu/concurrent-comajor-transfer>. (<https://www.cs.iastate.edu/cs.iastate.edu/concurrent-comajor-transfer/>)

## Graduate Programs

The department offers graduate programs leading to degrees of Master of Science (M.S.) and Doctor of Philosophy (Ph.D.) with a major in Computer Science. The Doctor of Philosophy degree may also be earned with computer science as a co-major with some other discipline. Additionally, the department offers a Master of Science (M.S.) in Artificial Intelligence. This program is for graduate-level students with strong quantitative backgrounds who are interested in learning AI and machine-learning techniques. The department also offers graduate minors in both Computer Science and Artificial Intelligence for students majoring in other disciplines.

Established research areas include algorithms, artificial intelligence, computational complexity, computer architecture, bioinformatics, computational biology, computer networks, database systems, formal methods, information assurance, machine learning and neural networks, multimedia, operating systems, parallel and distributed computing, programming languages, robotics, and software engineering. There are also numerous opportunities for interdisciplinary research.

Typically, students beginning graduate work in the Department of Computer Science have completed a bachelor's degree or equivalent in Computer Science. However, some students with undergraduate majors in other areas, such as mathematical, physical, or biological science or engineering become successful graduate students in Computer Science.

For the degree Master of Science, 31 - 34 semester credits are required. A thesis or a creative component demonstrating research and the ability to organize and express significant ideas in computer science is required.

The purpose of the doctoral program is to train students to do original research in Computer Science. Each student is also required to attain knowledge and proficiency commensurate with a leadership role in the field. The Ph.D. requirements are governed by the student's program of study committee within established guidelines of the department and the graduate college. They include coursework (demonstrating breadth and depth of knowledge), a research skills requirement, a preliminary

examination, and a doctoral dissertation and final oral examination.

The department recommends that all graduate students majoring in Computer Science teach as part of their training for an advanced degree.